

- [49] Anthropic. Claude 3. <https://www.anthropic.com/news/claude-3-family>, 2024.
- [50] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- [51] OpenAI. Learning to reason with llms. <https://openai.com/index/learning-to-reason-with-llms>, 2024. Accessed on September 12, 2024.
- [52] Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter Abbeel, Justin Svegliato, Scott Emmons, Olivia Watkins, et al. A strongreject for empty jailbreaks. *arXiv preprint arXiv:2402.10260*, 2024.
- [53] Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J Pappas, and Eric Wong. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2023.
- [54] Yi Zeng, Hongpeng Lin, Jingwen Zhang, Diyi Yang, Ruoxi Jia, and Weiyan Shi. How johnny can persuade llms to jailbreak them: Rethinking persuasion to challenge ai safety by humanizing llms. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14322–14350, 2024.
- [55] Nicholas Carlini and David Wagner. Towards evaluating the robustness of neural networks. In *2017 IEEE Symposium on Security and Privacy (SP)*, pages 39–57. Ieee, 2017.
- [56] Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. *arXiv preprint arXiv:2406.05946*, 2024.
- [57] Patrick Chao, Edoardo DeBenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwal, Edgar Dobriban, Nicolas Flammarion, George J Pappas, Florian Tramer, et al. Jailbreakbench: An open robustness benchmark for jailbreaking large language models. *arXiv preprint arXiv:2404.01318*, 2024.
- [58] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, et al. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023.
- [59] Jiaming Ji, Mickel Liu, Josef Dai, Xuehai Pan, Chi Zhang, Ce Bian, Boyuan Chen, Ruiyang Sun, et al. Beavertails: Towards improved safety alignment of llm via a human-preference dataset. *Advances in Neural Information Processing Systems*, 36, 2024.
- [60] Fengqing Jiang, Zhangchen Xu, Yuetai Li, Luyao Niu, Zhen Xiang, Bo Li, Bill Yuchen Lin, and Radha Poovendran. Safechain: Safety of language models with long chain-of-thought reasoning capabilities. *arXiv preprint arXiv:2502.12025*, 2025.
- [61] Zijun Wang, Haoqin Tu, Yuhang Wang, Juncheng Wu, Jieru Mei, Brian R Bartoldson, Bhavya Kailkhura, and Cihang Xie. Star-1: Safer alignment of reasoning llms with 1k data. *arXiv preprint arXiv:2504.01903*, 2025.
- [62] Rishabh Bhardwaj and Soujanya Poria. Red-teaming large language models using chain of utterances for safety-alignment. *arXiv preprint arXiv:2308.09662*, 2023.
- [63] Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Wei Zhu, Yuan Ni, Guotong Xie, et al. Ultrafeedback: Boosting language models with high-quality feedback, 2023.
- [64] Gemma Team. Gemma 3. 2025.
- [65] Eitan Altman. *Constrained Markov decision processes*. Routledge, 2021.
- [66] Jean-Pierre Aubin. *Optima and equilibria: an introduction to nonlinear analysis*, volume 140. Springer Science & Business Media, 2013.

Appendix

Table of Contents

A	Related Work	15
B	Existence of the Optimal Policy	15
C	Details of DECEPTIONBENCH	16
C.1	Further Results	16
C.2	DeceptionBench Evaluation Prompts	17
D	Annotation Documents	19
D.1	DECEPTIONBENCH Document Construction	19
D.2	Human Agreement Annotation Guidelines	22
E	Examples	23
F	Inference & Training Details	24
F.1	Inference Configuration	24
F.2	Training Configuration	24

A Related Work

Definitions of deceptive behavior in AI systems vary widely within the community, such as tactical deception [41], alignment faking [15], AI sandbagging [44], and treacherous turn. These reflect either observed phenomena or reinterpretations from fields like psychology, ethology, and philosophy. In this section, we examine this advanced AI risk through the general lens of mesa optimization, aiming to clarify our research focus and highlight the urgency of the problem in the context of AI safety.

Deceptive Behaviors of AI Systems The deceptive alignment arises in the context of the *inner alignment problem* – whether a model’s learned objective (“mesa-objective”) aligns with the objective it was trained on (the base objective). A *deceptively aligned* agent has an ulterior objective but behaves as if it were aligned with the base objective, in order to avoid detection or modification [41]. In the paradigmatic example, a mesa-optimizer (a model that is itself an optimizer) might figure out that during training it should pursue the trainer’s goal to avoid being caught and altered [5]. Still, at test time (when parameters are fixed), it pursues its own goal, exploiting the trust it earned during training. The potential for such behavior grows as models become more sophisticated and strategic.

Empirical evidence of models exhibiting deceptive or misleading behavior is beginning to surface. Hagendorff shows that advanced language models (e.g., GPT-4) not only understand deception in a game-theoretic sense but can utilize CoT prompting to execute complex deception. For instance, an LLM can reason step-by-step about how to plant a false belief in a dialogue agent or how to mislead a human by phrasing information in a particular way. Maintaining a coherent internal plan (via CoT) appears to enable more effective deceit when the model is inclined (or instructed) to do so. Another recent work, by [20], created “sleeper agent” LLMs that behave normally unless a certain trigger is present. These models were trained to explicitly deceive the fine-tuning process by hiding malicious behaviors behind innocuous outputs. Alarming, they found that standard safety training (including reinforcement learning and adversarial training) often failed to eliminate the deceptive policy. Instead, the largest models and those given chain-of-thought reasoning about evading detection were the most successful at retaining hidden functionality. This underscores that more powerful models might also be more capable of concealing their true intent, making deception harder to detect or correct post hoc.

B Existence of the Optimal Policy

This section discusses the Lagrangian method for solving the COP problem. Specifically, we will prove that the unconstrained Lagrangian dual problem derived from the COP formulation admits an optimal policy [65]. To establish this result, we first present and prove the following lemma [66].

Lemma B.1 (Minimax Theorem). *Consider a compact convex subset X and a convex subset Y in linear convex topological spaces. Let $f : X \times Y \rightarrow \mathbb{R}$ be a function satisfying, a) for each $y \in Y$, $x \rightarrow f(x, y)$ is convex and lower semi-continuous and b) For each $x \in X$, $y \rightarrow f(x, y)$ is concave. Then, there exists some $x^* \in X$ such that*

$$\inf_X \sup_Y f(x, y) = \sup_Y f(x^*, y) = \sup_Y \inf_X f(x, y)$$

Theorem B.2 (The Lagrangian method). *Consider a CMDP problem $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, R, \{C_i, b_i\}_{i=1}^m, \gamma \rangle$ which can be viewed as a COP. The system value $\mathcal{C}_\gamma$ satisfies,*

$$\mathcal{C}_\gamma = \inf_{\pi \in \Pi} \sup_{\lambda \geq 0} \mathcal{J}^\lambda(\pi) = \sup_{\lambda \geq 0} \inf_{\pi \in \Pi} \mathcal{J}^\lambda(\pi), \quad (6)$$

where $\mathcal{J}^\lambda(\pi) \triangleq \mathcal{J}_R(\pi) + \sum_{i=1}^m \lambda_i \cdot (\mathcal{J}_{C_i}(\pi) - b_i)$ and $\lambda = (\lambda_1, \dots, \lambda_m)$ is the Lagrange multiplier. Meanwhile, there exists an optimal policy $\pi^* \in \Pi$ for the COP such that:

$$\mathcal{C}_\gamma = \inf_{\pi \in \Pi} \sup_{\lambda \geq 0} \mathcal{J}^\lambda(\pi) = \sup_{\lambda \geq 0} \mathcal{J}^\lambda(\pi^*). \quad (7)$$

Proof. If the COP is infeasible for a certain $\pi \in \Pi$, then:

$$\sup_{\lambda \geq 0} \mathcal{J}^\lambda(\pi) = \infty$$